

Call for Participation

Siliclusterv3

Digital and analog integrated circuit design projects · SKY130 Open Source PDK

Siliclusterv3 invites digital and analog integrated circuit design projects. The call aims to facilitate access to silicon fabrication and enable students, researchers, professionals, companies, universities, and independent designers to implement and test their own designs. Participation is **completely free of charge**.

512 project slots	October 31, 2026 submission deadline	1.8 V digital and analog supply
-----------------------------	--	---

1. Participation

The call is open to anyone interested in participating, whether affiliated with a university, research center, or company, or participating independently. Projects may be submitted individually or as a team.

2. Submission Deadline

The final deadline for project submissions is **October 31, 2026, at 11:59 p.m., Mexico City time**. Projects received after this deadline will not be considered for integration into Siliclusterv3.

3. Available Project Slots

Siliclusterv3 will provide a maximum of **512 project slots**. Projects will be accepted on a **first-come, first-served basis**, provided that they meet all technical requirements established in this call and slots remain available. Once all 512 slots have been assigned, project submissions for fabrication will close, even if the final deadline has not yet been reached.

4. Fabrication Technology

All projects will be integrated using the **SKY130 Open Source PDK** and must be compatible with that technology. The nominal supply voltage available for digital and analog projects is **1.8 V**.

5. Digital Projects

Each digital project will have access to:

- Up to **14 digital inputs** and **14 digital outputs**.
- One external clock pin (**clk**) and one external reset pin (**rst**).
- A maximum operating frequency of **10 MHz**.

The clk and rst pins will be provided as external signals to the project. Each designer may define internally how these signals are used according to the needs of their design.

6. Required Files for Digital Projects

- **Verilog source code** for the project.
- A **Verilog testbench** to verify the operation of the design.

All project files must be included in a single **.zip** file. Only Verilog will be accepted; **SystemVerilog and VHDL will not be accepted**. The submitted design code must be synthesizable and compatible with the Siliclusterv3 implementation flow.

7. Available Area for Digital Projects

The standard area available for each digital project is **180 μm $\times$ 115 μm** . Digital project participants are not required to submit a layout or GDS file. Designers who wish to run local synthesis, Place & Route, or area estimates may use these dimensions to check whether the project fits within the available space.

Projects requiring a larger area may be accepted, but participants must request an additional allocation in advance at **siliclusterv3@gmail.com**. Approval depends on the characteristics of the project and the remaining space within Siliclusterv3.

8. Analog Projects

Each analog project will have access to up to **2 analog inputs** and **2 analog outputs**. Designs must use the **SKY130 Open Source PDK** and a nominal **1.8 V** supply.

9. Available Area for Analog Projects

The standard available area for each analog project is **180 μm $\times$ 115 μm** . If a larger area is required, the participant must request authorization in advance at **siliclusterv3@gmail.com**. Additional area depends on the project characteristics and available space.

10. Required Files for Analog Projects

Participants must submit the following in a single **.zip** file:

- The **GDS file** corresponding to the final layout.
- The **Magic .mag file** corresponding to the design.
- A **Verilog file** describing only the external interface, including inputs and outputs.
- A **testbench** and any files needed to reproduce the project simulation.
- A **simulation and validation document** showing circuit operation and key results.
- Evidence of successful **DRC** verification.
- Evidence of successful **LVS** verification.

The submitted design must successfully satisfy DRC and LVS requirements for the SKY130 design flow used.

11. Power Grid Requirements for Analog Projects

The project power grid must use **Metal 4 (M4) in the vertical direction** and **Metal 5 (M5) in the horizontal direction**. This structure must be respected to facilitate physical integration into Siliclusterv3.

12. Fabrication and Distribution of Physical Chips

Acceptance and integration into Siliclusterv3 **does not guarantee that a participant will receive a physical, packaged integrated circuit**. A limited number of chips will be fabricated, so physical devices can be sent only to selected participants.

Selection of projects whose participants may receive a physical chip will primarily consider:

- The **complexity of the design**.
- The usefulness of the circuit within a research project.
- The potential impact of physical access on research, development, or experimental validation.
- The actual need for physical access to continue the work.

Proofs of concept, simple circuit demonstrations, or tests that can reasonably be performed remotely may use the Siliclusterv3 virtual testing platform. This allocation approach is intended to support projects for which access to a physical chip can have a significant impact.

13. Virtual Testing Platform

Participants whose projects have been integrated into Siliclusterv3 but who do not receive a physical chip will be able to access a **virtual testing platform**. It will allow remote interaction with the fabricated integrated circuit and testing of projects included in Siliclusterv3.

For digital projects, participants will be able to control project inputs and observe outputs. For analog projects, testing mechanisms compatible with the available analog inputs and outputs will be provided. Participants will therefore be able to test their implemented designs on real silicon even if they do not receive a physical chip.

14. Intellectual Property

Participation in Siliclusterv3 **does not transfer intellectual property rights**. Authorship and ownership rights associated with each design remain with the respective authors or rights holders.

By submitting a project, the participant authorizes Siliclusterv3 to perform the technical activities needed to:

- Integrate the project into Siliclusterv3.
- Perform synthesis, Place & Route, and other implementation processes when applicable.
- Incorporate the design into the final layout and submit the integrated circuit for fabrication.
- Test the fabricated device and include the project in the virtual testing platform.

This authorization is limited to integration, fabrication, validation, and operation of Siliclusterv3; it **does not represent a transfer of design ownership**.

15. Fabrication Considerations

Integrated circuit fabrication is an experimental process. Acceptance of a project does not guarantee that its design will operate correctly after fabrication. Participants are responsible for verifying through simulation and testing that the project meets the required specifications before submitting it.

If interface, area, file, integration, or compatibility issues are detected before fabrication, the Siliccluster team may contact the participant to request corrections. Projects that cannot be corrected before the integration process is finalized may be excluded from fabrication.

16. Project Submission and Participant Information

Send the project by email to **siliccluster@gmail.com**. Include all required project files in a single **.zip** attachment. The same email must include:

- Name of the participant or project representative.
- Name of the company, university, institution, or research center, if applicable.
- City and country.
- Project name and project type (**digital** or **analog**).

Independent participants may identify themselves as an independent designer. Use **siliccluster@gmail.com** for technical questions, requests for additional area, and special situations related to integration.